

Chimneys, Draft and Termination

Why a chimney has to be tall, warm and its own — three rules that explain almost every chimney question.

A chimney works on draft: hot gas is lighter than cold air, so it rises and pulls more air in behind it. Anything that cools the gas or disturbs the pressure at the top weakens or reverses that flow.

Height — the 3/2/10 rule

The numbers here

Chimney above the roof penetration	3 ft
Higher than anything within 10 ft, by	2 ft
Unsupported chimney above the roof — maximum	6 ft

Three feet above where it comes through the roof, and two feet higher than anything within ten feet of it. Three, two, ten.

The best reason a chimney should terminate above nearby obstructions is that combustion products may otherwise be forced down the flue by wind. Wind striking a nearby roof or wall creates a zone of higher pressure at the chimney top, and that pressure pushes the exhaust back down.

Warmth

YOU HAVE MISSED THIS 2 TIMES

A chimney should not be located on an exterior wall because:

The flue temperature may be too cold.

A chimney inside the building stays warm between fires, so it establishes draft immediately. One on an exterior wall is chilled on three sides, and a cold flue starts by pouring cold air down into the room instead of drawing smoke up. Cold flue also means more condensation and more creosote.

One fire, one flue

A fireplace may not share a flue with other appliances. The exception is a fireplace flue-insert, which is not another appliance but the same fireplace fitted with a liner. Sharing a flue lets one appliance's exhaust spill into the other's opening.

Getting through the roof and the chase

YOU HAVE MISSED THIS 2 TIMES

A flue that penetrates directly through a roof should be flashed like a:

Metal vent.

Factory-built metal chimney through a framed chase

Requires a metal pan or cap at the top of the chase, so water cannot run down inside the framing.

Direct vent fireplace

Obtains combustion air from outside the house and expels products of combustion outside the house. It is sealed from the room entirely, so it cannot compete with the house for air.

Say it back

Cover the page. Answer each one out loud.

1. Explain draft in one sentence.
2. State the 3/2/10 rule, and the maximum unsupported height.
3. Why must a chimney terminate above nearby obstructions?
4. Why is an exterior wall a poor place for a chimney?
5. May a fireplace share a flue? What is the exception?
6. How is a flue penetrating a roof flashed?
7. What does a metal chimney in a framed chase require?
8. What makes a direct vent fireplace different?